A decorative graphic on the left side of the slide, consisting of a black crosshair with a blue square in the top-left quadrant, a red square in the bottom-left quadrant, and a yellow square in the bottom-right quadrant.

Computer Programming II (CSE2035)

Week 5 Lab Session:

Object Oriented Programming Part I

Dept. of Computer Science and Engineering
Sogang University

Exercise 0: My String

- Let's write a string class that works like the C++ string
- Properties, constructor, and Print() are given in my-string.hpp file (**do not change this file**)

```
class MyString {
private:
    char *arr;
    size_t length;
public:
    MyString(const char *s) {
        length = strlen(s);
        if (length == 0) { arr = NULL; }
        else {
            arr = (char*) malloc(length);
            memcpy(arr, s, length);
        }
    }
}
```

Exercise 0: My String

- **Implement the following functions in my-string.cpp**
 - If an invalid 'idx' or 'n' argument is given in Insert() or Remove(), don't do anything and just return.

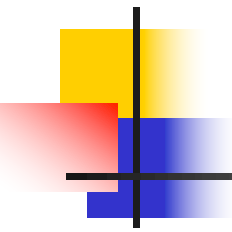
```
// Should not leave any heap-allocated memory.
~MyString();
// Get the length of the string.
size_t GetLength();
// Make the string into an empty string.
void Clear();
// Append 's' at the end of this string.
void Append(MyString *s);
// Insert 's' at position 'idx' of this string.
void Insert(MyString *s, int idx);
// Starting from the position 'idx', remove 'n' characters.
void Remove(int idx, int n);
```

Exercise 0: My String

- **Make sure that there is no memory leak**
 - After test() returns, there shouldn't be any heap-allocated memory that is not free()-ed yet.

```
void test() {
    MyString s1("ABC");
    MyString s2(" DEF");
    s1.Append(&s2);
    s1.Print(); // Should print "ABC DEF".
    cout << s1.GetLength() << endl; // Should print "7".
}

int main (void) {
    test();
    return 0;
}
```



Lab Assignment Submission Instructions

- **Create “WeekX_Lab_yourID_name” folder**
 - E.g., Week5_Lab_20221234_문의현
- **Put your completed works (e.g., codes/files/programs) to the folder**
- **Compress (Zip) your folder**
 - E.g., Week5_Lab_20221234_문의현.zip
- **Submit/Upload your Zip file to the Cyber Campus in the “Assignments” section**
- **Due dates: All the lab assignments are due by Wednesday at 11:59pm, right after prior week’s lab session on Thursday**
 - Late submissions are not accepted